Gait Disorder Assessment Based on a Large-Scale Clinical Trial: WiFi Versus Video Versus Doctor's Visual Inspection

Alireza Parsay, *Member, IEEE*, Mert Torun, *Member, IEEE*, Philip R. Delio, and Yasamin Mostofi, *Fellow, IEEE*

***Abstract*—Neurological gait disorders affect a large population, significantly reducing life quality. This article brings a foundational understanding to the potentials of emerging sensing modalities (e.g., WiFi) for gait disorder assessment, via conducting a one-year-long clinical trial in collaboration with the Neurology Associates of Santa Barbara. Our medical campaign encompasses 114 real subjects and a wide spectrum of disorders (e.g., Parkinson's, Neuropathy, Post Stroke, Dementia, Arthritis). We then develop the first WiFi-based gait disorder sensing system of its kind, distinguished by its scope of validation with a large and diverse patient cohort. To ensure generalizability, we mainly leverage publicly accessible online videos of gait disorders for training, and develop a video-to-RF pipeline to convert them to synthetic RF training data. We then extensively test the system in a neurology center (i.e., the Neurology Associates of Santa Barbara). Additionally, we provide a 1-1 comparison with a vision-based system, by developing a vision-based gait assessment system under identical conditions, a first-of-its-kind comparison to our knowledge. We finally contrast both systems with neurologists' accuracy when basing evaluation solely on visual gait inspection, by designing/distributing a large survey to 70 neurologists, offering the first apples-to-apples comparison of these three sensing modalities. Our findings can help integrate these sensing systems into medical practice, working toward equitable healthcare.**

***Index Terms*—Equitable healthcare, gait disorder, neurological disorder, RF sensing, smart health, WiFi sensing.**

I. Introduction

NEUROLOGICAL gait disorders affect 24% of adults ages 60 and older, with the number increasing to 55% above the age of 80 [1]. In general, gait disorders can be devastating to the mobility, independence, cognition, and self-esteem of an individual, as many studies have shown [2]. As such, proper diagnosis is crucial for giving the individual proper care that can optimize their well-being. Yet, many individuals may not seek medical help, or may seek it only after the disease has progressed [2]. Furthermore, once medical help is sought, many patients miss follow-up appointments [3]. The situation can be much worse in impoverished/developing nations (or even in rural areas within the United States) as families can face additional challenges in accessing healthcare due to prohibitive costs, extended waiting time for specialist consultations, or the general scarcity of medical expertise.

On the other hand, other promising sensing modalities have emerged over the past few years. For instance, wireless signals, now ubiquitous, have created new opportunities for sensing and learning about the environment, causing much enthusiasm in the area of RF sensing. More specifically, RF signals have been used for applications, such as person identification [4], [5], [6], [7], [8], crowd analytics [9], [10], [11], fall detection [12], [13], [14], and activity recognition [15], [16], [17], among others. Similarly, in the area of vision, the widespread availability of affordable, high-resolution cameras has opened up new sensing possibilities for various applications.

In this article, our first major goal is to fundamentally understand the potential of everyday WiFi signals for neurological gait disorder diagnosis. While RF-based gait disorder analysis has started to gain attraction in the literature, there are still very limited work in this area. For instance, existing WiFi-based ones either do not test with real patients or have only tested with a handful of patients, are limited in the range of considered disorders, and are typically trained and tested in the same environment (i.e., not in the wild) [18], [19]. On the other hand, there are a small number of papers using radar signals for gait disorder assessment [20], [21], [22]. However, the majority of these studies are also afflicted by the previous limitations. More importantly, such work use specialized equipment, which can be cost-prohibitive. In contrast, *a WiFi-based approach could offer a considerably more economically viable alternative, with the potential to democratize access to healthcare.* The next section provides a detailed comparison with the prior work, with the key points further summarized in Table I.

In the area of vision, on the other hand, there are more work on gait disorder assessment with cameras, due to the prevalence of video datasets that can be used for training/testing purposes. *Yet, there are no apples-to-apples comparisons between WiFi*

Received 21 July 2024; revised 24 October 2024 and 21 December 2024; accepted 21 February 2025. Date of publication 11 March 2025; date of current version 9 June 2025. This work was supported in part by NSF CNS under Award 2226255, and in part by the "Office of Naval Research (ONR) under Award N00014-23-1-2715. *(Alireza Parsay and Mert Torun contributed equally to this work.) (Corresponding author: Alireza Parsay.)*

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Institutional Review Board (IRB) Committee at UCSB.

Alireza Parsay, Mert Torun, and Yasamin Mostofi are with the Department of Electrical and Computer Engineering, University of California at Santa Barbara, Santa Barbara, CA 93106 USA (e-mail: alirezaparsay@ucsb.edu; merttorun@ucsb.edu; ymostofi@ece.ucsb.edu).

Philip R. Delio is with the Neurology Associates of Santa Barbara, Santa Barbara, CA 93105 USA, and also with the Comprehensive Stroke Center, Santa Barbara Cottage Hospital, Santa Barbara, CA 93105 USA (e-mail: prdelio@sbneuro.com).

Digital Object Identifier 10.1109/JIOT.2025.3550204

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and video-based systems— i.e., evaluations conducted under identical training and testing conditions and with the same subjects. Such a comparison is crucially needed for a proper integration of these systems in medical practices. Our second key objective is then to create a vision-based gait disorder system, mirroring the same subjects/conditions of our WiFi-based one, to facilitate a one-to-one performance comparison.

Finally, a successful integration of such sensing modalities in a smart healthcare system requires rigorous benchmarking against the diagnostic performance of medical experts. One key role the neurologists play is *the visual assessment of the gait*. Thus, the accurate duplication of this aspect by an RF-based or a vision-based system holds promises for revolutionizing the diagnostic procedure. In other words, neurologists utilize many inputs (e.g., blood tests, genetic tests, etc) to establish a diagnosis, in addition to a visual inspection of the gait. However, many of these inputs can be collected more easily (i.e., do not require the specialized expertise of neurologists) and integrated into a smart health system. Thus, if a smart sensing system can replicate the complex task of visual inspection, it has the potential to automate the diagnostic process, reducing the burden on specialists, and enabling healthcare access for a larger population.

However, there are no existing work on assessing the diagnostic accuracy of neurologists if only relying on the visual inspection of the gait, motivating our third goal. More specifically, we design a large survey and disseminate it to 70 board-certified practicing neurologists via partnership with survey healthcare global (SHG) [23], in order to collect their performance when providing a diagnosis only based on the visual inspection of the gait. This then enables an apples-to-apples comparison with both our WiFi-based and video-based systems.

We next explicitly discuss the contributions of this article, followed by a detailed state-of-the-art comparison.

Statement of Contributions:

1) Through a partnership with neurologists, we provide a comprehensive analysis of WiFi-based gait disorder assessment *by running a one-year-long clinical trial encompassing 114 subjects and a wide range of disorders*. These conditions include Parkinson's disease, Neuropathy, Post Stroke, Dementia, and Arthritis, in addition to healthy subjects. The subjects further exhibit different degrees of severity: mild, moderate, and severe. Finally, the testing is done in the wild, i.e., in the doctor's office. Beyond demonstrating the performance of WiFi signals when detecting gait disorders, this article further comprehensively shows the impact of several different parameters (e.g., age, height, weight and disease severity) on the performance. It additionally tests with a few samples of other conditions (amyotrophic lateral sclerosis (ALS), multiple sclerosis (MS), spinal cord injury, neurofibromatosis, and dysferlinopathy), as well as cases with multiple disorders. To the best of our knowledge, *this article serves as the first in-depth evaluation of WiFi's capability in gait disorder assessment*, with the potential to make a significant leap forward in the field.
2) To ensure generalizability of our findings and further alleviate the extensive data collection required for training purposes, *we have trained a neural network for RF-based gait disorder detection by mainly leveraging publicly accessible online videos of gait disorders/healthy gait*s. This is enabled by our video-to-RF pipeline that converts such videos of a person's walk to the corresponding synthetic RF data, as if the walk was performed near a pair of WiFi transceivers.
3) Equally important, the article provides a detailed one-to-one comparison with a vision-based system, by further *developing a vision-based gait disorder assessment system that uses identical subjects and conditions for both training and testing purpose*s. This side-by-side comparison is crucial toward developing smart health systems and the first of its kind to the best of our knowledge. Furthermore, as will be demonstrated, the findings indicate comparable performance between the two modalities.
4) In order to establish the diagnostic accuracy of neurologists when only using visual inspection of the gait, we have designed a large video-based survey (based on the same pool of data) and have further partnered with SHG, an entity that provides "measurable healthcare expert opinions," via recruitment and other tools, who then disseminated our survey to 70 board-certified practicing neurologists. We then contrast the performance of our WiFi-based and video-based systems with the diagnostic accuracy of neurologists when they base their evaluation solely on the visual inspection of the gait. *To the best of our knowledge, this is the first result of its kind, which provides an apples-to-apples comparison of these three sensing modalities*. As the results will show, both WiFi and Video-based systems surpass the performance of neurologists when only using visual inspection of the gait.

Overall, the findings of this article can provide guidelines for smart health system development, shape the integration of these new sensing modalities into medical practice, and pave the way toward equitable access to health. *We note that our IRB committee has reviewed and approved this research.*

Remark 1: Our WiFi CSI dataset, collected from patients with various gait disorders as well as from healthy individuals, is available on our website [24] or upon request to the authors.

Remark 2: For brevity, in the rest of this article, we refer to the Neurology Associates of Santa Barbara where the experiments took place as *The Neurology Center*.

II. Related Work

This section reviews emerging sensing systems for gait-based neurological disorders. It further extensively compares this work with state-of-the-art. A detailed summary of this comparison is also provided in Table I for RF-based systems.

The prior work can be classified to three main categories: 1) methods utilizing floor/wearable sensors; 2) video-based techniques; and 3) RF-based approaches.

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TABLE I
STATE-OF-THE-ART COMPARISON OF RF-BASED GAIT DISORDER ASSESSMENT. WE NOTE THAT WHILE WE HAVE STATED SIX CONDITIONS FOR THIS ARTICLE, WE FURTHER TEST WITH FIVE MORE CONDITIONS, BUT AT SMALLER SCALE

Ref.	Method	Scope of Evaluation	# of Conditions (incl. Healthy)	Training Data Collection Burden	Train/Test in Diff. Environment	Cost	Accuracy
[20]	Radar (Gait-based)	24 Real Subjects	2	High	No	High	94.9%
[21]	Radar (Gait-based)	Only acted out	5	High	No	High	88.4%
[22]	Radar (Breathing-based)	53 Real Subjects	2	Low	Yes	High	84.53%
[41]	WiFi (Endurance-based)	Only acted out	2	High	No	Low	99.4%
[19]	WiFi (Gait-based)	15 Real Subjects	1	High	No	Low	99.7%
[18]	WiFi (Gait-based)	Only acted out	7	High	Yes	Low	94%
This Paper	**WiFi (Gait-based)**	**114 Real Subjects**	6	Low	Yes	Low	**85.47%**

1) *Wearable and Floor Sensors:* Accelerometers, gyroscopes, and magnetometers have been recently used as floor/wearable sensors for classifying neurological disorders that have a gait manifestation [25], [26], [27], [28], [29], [30], [31], [32]. These sensors are typically attached to the body, e.g., smartwatches and insole pressure sensors, or installed on the ground/treadmill, e.g., force plates. Overall, while such recent work can be valuable in better understanding an abnormal gait pattern, incorporating them into practice comes with significant drawbacks, such as limited public accessibility (price and delivery), patient discomfort, and difficulty in use [33], [34].

2) *Video-Based Methods:* With the increasing accessibility and advancements in computer vision techniques, utilizing video footage for gait disorder assessment has become a recent area of exploration [33], [34], [35], [36], [37], [38], [39]. For instance, [35] develops an R-CNN-based classifier to distinguish between healthy and Parkinsonian gaits, while [37] performs classification (health versus unhealthy), based on 2-D pose estimation. Similarly, Verlekar et al. [38] use 2-D footage of actors, for gait impairment classification. Authors in [39] classify two gait-affecting neurodegenerative disorders: Parkinson's and Hemiplegia, by utilizing a 3-D Kinect sensor. On the other hand, [40] classifies healthy versus unhealthy gaits, considering five unhealthy gait disorders that are acted out by ten subjects.

In general, the video-based systems, while simpler-to-use as compared to wearables, can raise privacy concerns and/or may not be favorable for some patients that are not comfortable being camcorded. Furthermore, they require a line-of-sight view. *Nevertheless, in addition to advancing the state-of-the-art in the area of RF-based gait disorder assessment, this article further contributes to the area of vision-based gait assessment*, as discussed next.

In general, there are more papers on vision-based gait disorder assessment, as compared to RF-based approaches. However, a direct comparison between WiFi-based and vision-based systems, under identical conditions and with the same subjects, is currently lacking. Such a side-by-side methodical comparison is important for providing guidelines on how such systems can be incorporated into the future smart health ecosystem. This article thus contributes to the area of vision by providing such a comparison. Furthermore, this article contrasts the performance of a vision-based system (as well as its WiFi-based counterpart) with that of neurologists when they provide a diagnosis solely based on the visual evaluation of the gait.

3) *RF-Based Methods:* Radio frequency-based gait disorder analysis has recently gained attraction in the literature. However, *there are very limited number of work (six to the best of our knowledge) on gait-disorder assessment using RF signals*, as briefly summarized next. In [20], [21], IR-UWB and K-band radars are utilized to classify healthy versus abnormal gait, while [22] utilizes a customized frequency-modulated continuous wave (FMCW) radar to detect Parkinson's disease using nocturnal breathing signals. When considering WiFi signals, WiFreeze [19] aims to detect freeze of gait in Parkinson's patients, while [41] and [18] classify unhealthy versus healthy gaits. Less related to this work is Liu et al. [42], which does not detect gait disorders, but analyzes the progress of Parkinson's disease using customized FMCW radio equipment. We next compare our work with existing RF-based gait disorder studies, as also summarized in Table I.

Scope of Evaluation: Half of the existing RF-based work have not been tested with real patients, rather, the conditions have been enacted [18], [21], [41].

For those work that have tested with real patients [19], [20], [22], most collect a very small number of patients: 24 (only 4 unhealthy) in [20] and 15 in [19]. Reference [22] has collected data of 53 subjects, albeit with customized radar equipment, and for detection based on breathing signals. In contrast, our paper reports on a comprehensive one-year-long clinical trial involving WiFi-based sensing of 114 subjects, thereby offering a significantly broader scope of investigation.

Spectrum of Considered Gait Disorders: Most existing RF-based gait assessment work have only considered one type of gait disorder (e.g., healthy versus Parkinson). In this article, we have undertaken a comprehensive clinical campaign encompassing five common gait disorders: Parkinson's, arthritis, neuropathy, post-stroke, and dementia, plus healthy. In addition, we also test with the following conditions: ALS, MS, spinal cord injury, neurofibromatosis, and dysferlinopathy, but at a smaller scale.

Training Data Collection Burden: All the existing work (except for [22] which uses breathing signals and customized FMCW radar) have manually collected RF training data, which can be cumbersome and can result in nongeneralizable results.

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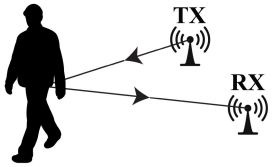


Fig. 1. High-level schematic of our WiFi-based gait disorder classification setup.

This work, however, mainly uses available online videos to generate synthetic RF data for training, thus significantly reducing the training data collection burden and further ensuring generalizable results.

Testing in the Wild: Most existing work (all WiFi-based ones) that collected data of real subjects train and test in the same environment (using the same overall pool), potentially yielding nongeneralizable results. In this article, on the other hand, we adopt a novel approach by training with synthetic RF data derived from online videos, while testing in The Neurology Center during its normal operating hours. This ensures the robustness and generalizability of the findings of this article.

Overall, this article provides a comprehensive analysis of RF-based gait disorder assessment by running a one-year-long clinical trial encompassing a large spectrum of disorders (five common gait disorders plus healthy). It then details the results of the data analysis of 114 subjects, while predominantly employing synthetic RF training data generated from available online videos, and relying only on off-the-shelf WiFi devices (a pair of laptops) for sensing. Equally important, this article provides a detailed comparison with our developed vision-based system, using identical subjects and conditions for both training and testing. It then contrasts the performance of these systems with the diagnostic accuracy of neurologists when they base their evaluation solely on the visual inspection of the gait. Overall, the findings of this article can provide guidelines for smart health system development and shape the integration of these new sensing modalities into medical practice, thus paving the way toward equitable healthcare.

III. Primer on RF Signal Modeling

Consider a person walking in an area, in the vicinity of a pair of WiFi transceivers, as shown in Fig. 1. The transmitter sends a signal, which bounces off of different parts of the body of the person, as well as other objects in the environment, and is received by the receiver. Then, the complex baseband received signal can be written as: $s_{\text{rec}}(t) = \alpha_s e^{j\theta_s} + \sum_{l\in V_{s'}} \alpha_{s',l} e^{j\theta_{s',l}} + \sum_{i\in V_b} \alpha_i e^{j((2\pi/\lambda)\psi \int v_i(t)dt + (2\pi/\lambda)d_i)}$, where λ is the wavelength [43]. Moreover, α_i is the magnitude of the scattered signal from the ith body part whose velocity is $v_i(t)$, and $\psi = \cos\phi_R + \cos\phi_T$, where ϕ_R and ϕ_T are the angles between the direction of motion and the vectors from the transmitter and receiver to the body, respectively. d_i is the initial length of the ith path, and V_b is the set of body points visible to both the receiver and the transmitter. The first and second terms then embrace the impact of the direct path and the reflections from static objects, respectively. We note that at WiFi frequencies, and for the area dimensions considered in this article, we have a far-field scenario. As such, we took ψ to be the same for all the body parts, as done in the literature. The WiFi receiver, however, can only measure the received signal magnitude or phase difference. We then next summarize an approximated modeling of the received RF signal magnitude [43], [44]. More specifically, since the LOS path is much stronger than the other reflections, i.e., $|\alpha_s| \gg |\alpha_i|$ and $|\alpha_s| \gg |\alpha_{s',l}|$, for $\forall i, l$, we have the following for the received signal magnitude

$$|s_{\text{rec}}(t)| \approx \gamma + \sum_{i\in V_b} \alpha_i \cos\left(\frac{2\pi}{\lambda}\psi \int v_i(t)dt + \frac{2\pi}{\lambda}d_i - \theta_s\right). \quad (1)$$

The first term, $\gamma = \alpha_s + \sum_{l\in V_{s'}} \alpha_{s',l} \cos(\theta_{s',l} - \theta_s)$, is static and removed after DC removal, and the second term embraces the information of the moving body parts. A similar expression can be derived for the phase difference signal.

IV. Synthetic RF Data Generation

A proper training of a classifier requires extensive training data. For WiFi sensing applications, however, WiFi sensing datasets pertaining to specific tasks of interest (e.g., gait disorder classification in this article) are not readily available. Furthermore, manual collection of such datasets can be prohibitive. In the area of vision, on the other hand, there is an abundance of vision dataset pertaining to many different scenarios. Thus, in recent years, there has been an interest in translating such vision datasets to synthetic RF training datasets (albeit for other applications) [17], [44], [45].

In this article, we then utilize a publicly available gait disorder vision dataset of real subjects for generating synthetic RF training data. In this section, we set forth the details of our pipeline for translating a vision input to a synthetic RF signal. As we shall see, the typical human mesh recovery approaches in vision result in a 3-D mesh that has a higher sampling rate for certain body parts. This can then magnify the impact of those body parts when generating synthetic RF signals. As such, in this part, we also propose a new approach for remedying this issue. We next discuss the steps in detail.

A. Human Mesh Recovery

Consider a video of a person walking with a gait disorder. We first utilize a state-of-the-art human mesh recovery algorithm (HMR) to extract a 3-D point cloud of the body for each frame [46]. However, the output 3-D mesh generated by HMR is a nonuniform point cloud whose spatial density is a function of the curvature of the body. Specifically, regions with higher surface curvature, such as the hands, are assigned a denser distribution of mesh points compared to flatter areas like the torso. This distribution is a feature of the SMPL (skinned multiperson linear) model [47], which serves as the base model in HMR.

Fig. 3 (a) shows an example of this where we have utilized state-of-the-art HMR to translate a snapshot of a walk into its 3-D point cloud. We can see the zoomed-in mesh points for two body parts: the hands and the legs, where hands are sampled much more heavily due to the high surface curvature.

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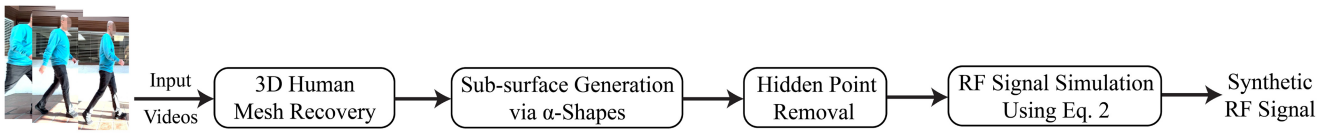


Fig. 2. Video-to-RF pipeline to generate synthetic RF training data from public gait disorder videos.

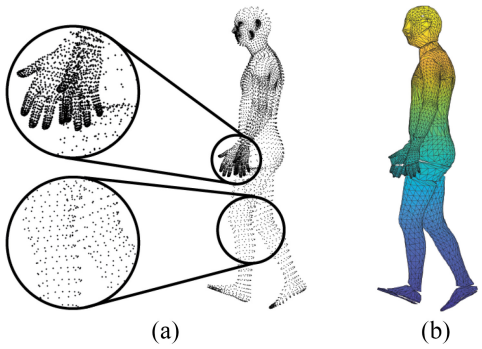


Fig. 3. Output of (a) HMR and its nonuniform sampling issue and (b) α-shape algorithm resolving it using Delaunay triangulation.

This can then result in the body parts with higher curvatures contributing more than they truly would to the received signal. We next show how to address this issue using α-shapes.

B. Finding α-Shapes of Human Mesh

HMR provides the 3-D mesh as a set of discrete point cloud. We then translate the points to nonoverlapping subsurfaces that cover the whole body using α-shapes. The α-shape of a set of points is a polytope (not necessarily convex) that best fits the corresponding set [48]. It can thus be considered as a generalization of convex hulls. By forming the α-shape of our given mesh points, we then effectively smooth out the impact of nonuniform sampling and further generate a set of nonoverlapping triangular subsurfaces, for wave simulation purposes. More specifically, we use the output body partitions provided by HMR (e.g., torso, shoulder, etc.) and create the α-shape of each body part, using Delaunay triangulation. Fig. 3 (b) shows the generated α-shapes for Fig. 3 (a). As seen, we now have a set of subsurfaces whose area is proportional to the area of the corresponding body part. By incorporating the area of each subsurface into the scattering coefficient as will be discussed in Section IV-E, we can effectively resolve the original issue caused by the nonuniform sampling rate.

C. Coordinate System Alignment

Since HMR provides its output in pixels, the α-shapes will also be in pixels. However, for synthetic wave simulation, the positions need to be converted to real-world coordinates. Therefore, by using the provided camera location and walk distance from the video, we then translate the pixels into meters. We further adjust for the angle between the camera and the body centroid at each point along the walking line to address the distortion caused by the viewing angle, similar to [44]. Finally, we place the resulting 3-D human shape, consisting of triangular subsurfaces, in the vicinity of a pair of transceivers in a simulation environment, according to the desired setup, to simulate the walking away motion [44].

D. Visibility Assessment

Let c_i and n_i denote the center and normal vector of the ith subsurface. We apply a visibility detection algorithm [49] to the centers to find subsurfaces visible to both the transmitter and receiver. With a finite sampling rate, this algorithm may misclassify a small set of points [49]. To address this, we perform an additional step. Let $\hat{x}_{\text{inc}}^{(i)}$ and $\hat{x}_{\text{RX}}^{(i)}$ denote the unit vector from the transmitter to c_i, and from c_i to the receiver, respectively. If $\langle \hat{x}_{\text{inc}}^{(i)}, n_i \rangle$ or $-\langle \hat{x}_{\text{RX}}^{(i)}, n_i \rangle$ is positive, then the ith point will not be visible to the transceivers, where $\langle , \rangle$ denotes the inner product. We then remove all such points if still in the output, thus further reducing the chance of false positives. We denote the final visible set by V_b.

E. Modeling the Scattering Pattern

Consider the specular (mirror-like) vector of the ith subsurface: $\hat{x}_{spec}^{(i)} = \hat{x}_{\text{inc}}^{(i)} - 2\langle \hat{x}_{\text{inc}}, n_i \rangle n_i$. We model the quasi-specular scattering pattern of the body using a Lambertian model [50], [51], $R(c_i) = \sqrt{A_i \cos(\theta_{\text{inc}}^{(i)})}(\alpha e^{j\phi} + \beta \cos^m(\theta_i))$, where α and β are the relative strength of these terms, m determines the directivity of the quasi-specular component, and ϕ is randomly chosen from $[0, 2\pi]$. Moreover, A_i is the area of the ith subsurface and θ_i is the angle between $\hat{x}_{\text{RX}}^{(i)}$ and $\hat{x}_{spec}^{(i)}$. In this manner, $A_i \cos(\theta_{\text{inc}}^{(i)})$ takes the effective area of the ith surface into account. This further ensures that body surfaces with a higher curvature, which inherently have more sampling points assigned to them (as discussed in Section IV-A), do not unrealistically affect the total received power.

F. Synthetic Wave Simulation

We finally simulate the interaction of the incoming wave with the 3-D human shape consisting of the α-shape-based subsurfaces of Section IV-B, with the transceivers as discussed in Section IV-C, and by using the visibility assessment and scattering model of Sections IV-D and IV-E, respectively. More specifically, by using Born approximation, we have [43], [44]

$$s_r = g(P_{TX}, P_{\text{RX}}) + \sum_{i \in V_b} g(P_{TX}, c_i) R(c_i) g(c_i, P_{\text{RX}}) \quad (2)$$

where $g(p_1, p_2) = [1/(4\pi \|p_1 - p_2\|)] e^{j[(2\pi \|p_1 - p_2\|)/\lambda]}$ is the free space Green's function, $R(c_i)$ is the scattering coefficient of the ith subsurface discussed in Section IV-E, and V_b is the set of visible points obtained in Section IV-D. Eq. (2) then allows us to translate a video input to synthetic RF data, which we shall use for training purposes in this article.

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V. Clinical Trial of WiFi-based Gait-Disorder Assessment: Setup

In this section, we present the details of our one-year-long clinical trial for testing the proposed pipeline. This is then followed by Section VI, where we extensively show the performance of the proposed pipeline when assessing the gait in The Neurology Center.

A. Experiment Environment

We have partnered with board-certified neurologists from The Neurology Center, in order to extensively test our proposed pipeline with real patients exhibiting a variety of gait disorders, and in a real office space. Through this partnership, the clinic dedicated a 7m $\times$ 3.18m covered space to our experiments, as shown in Fig. 4. We have then placed a pair of laptops in this area, where one acts as a WiFi transmitter and the other as a WiFi receiver.

As discussed in Section I, the system is envisioned to be used in smart health settings (e.g., smart home setting). As such, patients should be asked to walk the most meaningful route for the assessment, which is the one perpendicular to the line connecting the transmitter and receiver. Then, the patients are asked to walk back and forth two times, starting from the marked start point to the marked end point. We note that we do not have any control over how the patients walk, and as such, they can deviate from a straight path at times. We finally note that the route is *not marked* on the ground from the start to the endpoint, as enforcing walking on a marked route can affect how the patients walk, and can thus obscure the gait disorder. We then place the transmitter and receiver 64 cm apart (std=16 cm), and at the height of 1.1 m (std=5 cm). The distance from the start point to the link connecting the transmitter and receiver is 95 cm (std=15 cm). Then, the patients are asked to walk back and forth two times, from the start point to the end point, where each one-way is 3.9 m (std=0.1 m). We note that the observed slight variability in these parameters is due to the restrictions from The Neurology Center on leaving behind floor markers/equipment.

Remark 3: In this article, a gait sample refers to a one-way walk from start to end in Fig. 4, or vice versa.

B. Duration of the Clinical Trial

We have run extensive experimental validations with real subjects in the aforementioned Neurology Center over the course of one year. Since some patients may revisit the clinic, we have ensured that each patient is only included once in our experiments.

C. Test Subjects

In the first phase of this clinical trial, we have tested our proposed pipeline with the WiFi data of 101 subjects, spanning over 6 gait conditions: Parkinson's disease (20), dementia (14), post-stroke (16), arthritis (hip or knee, 8), neuropathy (10), and healthy (33). We note that the uneven distribution of the numbers across conditions partly mirrors the visit frequency of each class to the clinic.



Fig. 4. Two views of our clinical trial area in the neurology center.

We further note that the participants may experience different degrees of disease severity, which can affect the sensing performance. We extensively discuss such impacts in the next section. Finally, we note that we have also collected additional data from subjects suffering from other conditions, or multiple simultaneous conditions, as detailed in the next section.

It is important to collect the data of healthy subjects in the same environment for consistency. As such, the healthy individuals are recruited from the following groups: patients who visited the clinic for other neurological issues that do not have a gait manifestation (mainly from migraine patients), the companions of the patients, and the staff of The Neurology Center, among others.

As mentioned earlier, most patients walked back and forth two times. We then only use those parts where the participant walks away from the link as it is indicated to have a better signal quality [43]. This then amounts to two gait samples for most patients, each of which then independently contributes to the overall test pool. Overall, 101 subjects amounted to a total of 204 gait samples.

D. Human-Subject Protocol and Recruitment Process

During their visit to the center, the patients are handed out a pamphlet that details the purpose of the study, the procedure, the compensation, and other details to ensure compliance with the guidelines of our Institutional Review Board (IRB). Once a patient agrees to take part in our medical trials, he/she then walks back and forth a couple of times in the area of Fig. 4, as detailed in Section V-A. The collected data are then stored in an anonymized manner (using an identification code), and the ground-truth gait diagnosis is provided to us by the doctor.

We note that our IRB committee has reviewed and approved this research.

E. Establishing the Ground-Truth

Having a ground-truth for establishing the accuracy of our proposed system is crucial. Neurologists utilize many inputs to establish a diagnosis, in addition to a visual inspection. These can include: the results from vision and speech assessment, sensory perception and mental status evaluations, brain scans, genetic and blood tests, cerebro-spinal fluid analysis, and electroencephalography [52]. Our medical partners then furnish us with their definitive diagnosis for each patient, which we then adopt as the ground-truth. In addition, the doctor provide us with their assessment of the severity of each condition, categorizing it as mild, moderate, or severe.

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F. WiFi Data Collection and Processing

We use a pair of laptops equipped with an Intel 5300 WLAN card as transceivers to collect the WiFi data (See Fig. 4). While the patient is walking, the transmitter sends 500 packets/second in channel 64 with a center frequency of 5.32 GHz. For each of the three receiving antennas, the channel state information (CSI) of 30 subcarriers and the subsequent signal magnitude is obtained using CSI Tool [53], resulting in 90 data streams. We then denoise the data by applying principal component analysis (PCA). For each PCA component, a spectrogram is generated using a combination of multiwindow Hermite functions and STFT, based on a window of size 0.3 s and a time shift of 16 ms [43]. The final spectrogram is then generated by averaging over the first 15 spectrograms.

G. Generating Synthetic RF Training Data From Public Videos

In this article, we utilize available online videos to generate synthetic RF training data, as proposed in Section IV. More specifically, using available online videos [54], we have formed a video dataset of 78 subjects for synthetic WiFi training data generation, encompassing the following three gait conditions: 1) Parkinson's; 2) arthritis; and 3) healthy. This dataset has side-view perspective of the subjects, which is most informative for HMR [46]. We note that there are no publicly available dataset for the other three gait disorders of interest to this article. However, this has little impact on our performance as we shall see in the next section. Each subject then contributes to two gait samples (except for one person), amounting to a total video dataset of 155 gait samples: Parkinson's (26), arthritis (70), and healthy (59).

We then apply the proposed pipeline of Fig. 2 to each video. More specifically, the Human Mesh Recovery (HMR) algorithm of [46] is first applied to the video to extract a 3-D point cloud of the person, which is then up-sampled to 250 fps. We then apply the α-shape algorithm [48] to each body part individually, to generate the subsurfaces. For each body part, we set a fixed α-radius, which we empirically optimize by utilizing a small number of videos. In general, the smaller the body part, the smaller the α-shape radius should be.

After generating the subsurfaces, we set the WiFi transceivers in a simulation environment, according to the configuration of Fig. 4, in order to simulate the synthetic RF data that would have been generated if the person in the video was in a WiFi area. We then apply the wave simulator of Section IV, using the reflection model of Section IV-E and visibility assessment of Section IV-D. To empirically optimize and set the parameters for m, α, and β of this simulator, we simultaneously collect the WiFi and video data of two participants, one with normal and one with an abnormal gait, which are then only used for this purpose (not in training or testing). Overall, the end-to-end pipeline results in 155 synthetic RF gait samples of 78 subjects.

H. Spectrogram Generation and Feature Extraction

For each RF training data sample, we generate its spectrogram using a combination of multiwindow Hermite functions and STFT [43]. We note that we only consider the central 50% time span of each walk, in order to remove the acceleration and deceleration parts as well as any other unrelated movements at the onset or conclusion of the activity. We then extract six features from each spectrogram for training a machine learning pipeline, as discussed next.

In order to find the torso speed, we first sum the spectrogram over frequency for each time instant. We next obtain the shortest frequency window where the sum of the spectrogram values is greater than 50% of the overall sum (for each time step). This leads to two distinct curves that mark the lower and upper boundaries of the resultant frequency band. We then remove the outliers using the generalized extreme studentized deviate (GESD) test, and smooth the curves by running a moving average over four samples. Finally, we apply the same 50% method to the obtained spectrogram band and average the upper and lower bound curves to get the torso speed. We then find the following six features accordingly: 1) Average torso speed: This is the average of the torso speed over time; 2-3) Minimum and maximum of torso speed: We calculate the minimum and maximum speed of the torso from its speed curve; 3) Gait cycle: This is obtained by finding the peak values (at least 0.3 s apart) in the autocorrelation function of the torso speed [4], [44]; 4) Step length: We calculate the average step length, which is the stride time (half of gait cycle) times the average torso speed [43]; 5) Torso speed variations: We obtain the average difference between the maximum torso speed (90% percentile) and minimum torso speed (10% percentile) across all steps [43].

I. Training Gait Disorder Classifier for WiFi-Based Sensing

We train a feed-forward, fully connected multilayer perceptron (MLP) neural network with two hidden layers for WiFi-based gait disorder detection, given the features of Section V-H, and by using the generated synthetic video-to-RF data of 78 subjects. The hidden layers have 512 and 256 nodes, respectively. The dropout layers are applied to both hidden layers with the parameter 0.5. We utilize ReLU as the activation function for the hidden layers and softmax function at the output layer. Adam optimizer with a learning rate of 0.00001 and a weighted cross-entropy loss function is then used to train the network for 100 epochs, with a batch size of 10.

J. Domain Adaptation

The synthetic RF training data is generated from videos. Thus, it not only is a completely different modality, but it also naturally encompasses different subjects and environments as compared to those used in testing. These two factors result in the well-known domain and distribution shift problem between the training and test samples in machine learning [55]. To address this, we use WiFi data of 12 subjects (6 healthy, 3 Parkinson's, and 3 arthritis) to reweigh the loss function via an exhaustive search that aims to maximize the overall system's average accuracy for this small dataset. We then repeat the

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search process ten times and declare the median of the weight as the final value. To ensure robustness, the gait assessment's overall performance during testing is then averaged over 8 random sampling of the 12 domain-adaptation subjects, out of the collected WiFi data of 101 subjects, while the rest 89 subjects are left for testing in each round.

We next extensively discuss the performance of the proposed pipeline with real patients.

VI. Clinical Trial of WiFi-Based Gait-Disorder Assessment: Performance Evaluation

In this section, we present and analyze the outcomes of our one-year-long clinical trial in The Neurology Center, involving 114 subjects. Our analysis is comprehensive as it covers various different gait conditions. We further show the impact of the severity of the disease on the classification accuracy, in addition to demonstrating the impact of age, height, and weight on WiFi-based sensing of gait disorder. Moreover, we show the performance of our system with other gait conditions, such as ALS, spinal cord injury, dysferlinopathy, and MS, and demonstrate how they are mainly classified as unhealthy. We finally show the results when testing with patients that suffer from multiple gait conditions.

Overall, the results shed light on the feasibility of these signals for detecting gait disorders and can provide guidance on how they can be incorporated into future health systems.

A. Overall Performance

In this part, we discuss the performance of our classifier when tested with 101 subjects. More specifically, the overall average per subject accuracy of this classifier is 85.61%, with an average per class accuracy of 85.47%, with the following breakdown over the two classes: unhealthy (85.8%) and healthy (85.13%). Fig. 5 further shows the average classification accuracy per condition. As can be seen, the performance is mostly consistent per condition. The lower performance for some conditions (e.g., Parkinson's) can be largely attributed to the disparate distribution of disease severity across conditions, or to how differently these conditions manifest in gait.

B. Impact of Disease Severity

The severity of a gait disorder can play a key role in its identification, with the expectation that more severe cases have a more clear gait manifestation, making it easier to classify them. We next show the impact of disease severity on the classification accuracy. To establish ground truth, The Neurology Center has categorized each case as *mild, moderate, or severe*. Table II subsequently presents the breakdown of the probability of correct classification (as *unhealthy*) based on disease severity, for the unhealthy gait conditions. As can be seen, the accuracy increases as the severity of the disease increases, as expected.

C. Impact of Age, Weight, and Height

This section provides a demographic breakdown of the performance of the proposed classifier, across different age groups, genders, body weights, and heights. As seen in Fig. 6(a) and (b), the system shows a consistent performance over different ranges of heights and weights. Similarly, the classifier attained an accuracy of 86.61% for individuals self-identifying as males and 84.83% for those self-identifying as females, demonstrating its capability to accurately classify participants across gender lines. On the other hand, Fig. 6(c) illustrates enhanced performance in classifying older or younger participants, with a notable decline in the accuracy for middle-aged individuals. We can attribute the observed pattern to the fact that the older subjects typically exhibit more severe symptoms, while the younger group predominantly constitute healthy cases. Meanwhile, the middle-aged cohort generally exhibit symptoms that are less severe than those of older patients, while having a higher incidence of unhealthy cases than the younger group.

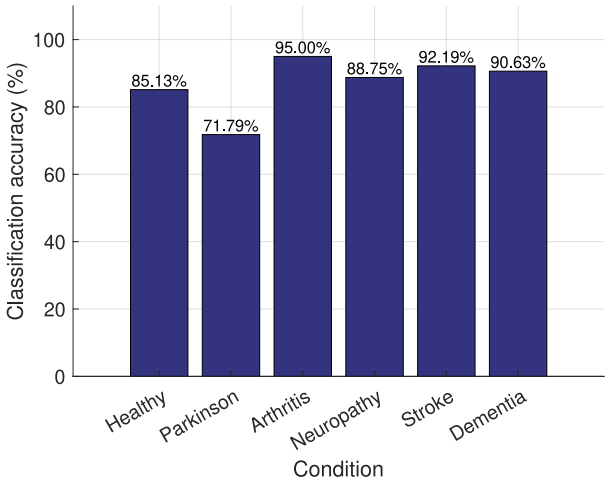


Fig. 5. Performance of our WiFi-based system in correctly classifying unhealthy/healthy gaits of 101 subjects in The Neurology Center, with a performance breakdown over each condition. It can be seen that the system can robustly classify unhealthy/healthy gaits when tested extensively with real subjects, with an overall per subject accuracy of 85.61%, although the system is mainly trained on synthetic data generated from online videos.

TABLE II
Impact of Disease Severity on Correctly Classifying It As Unhealthy. The Table Shows the Average Overall Performance, Averaged Over All the 5 Gait Disorder Conditions, As Function of Disease Severity

Severity	Mild	Moderate	Severe
Accuracy	59.38%	80.98%	100.00%

D. Other Gait Conditions

We next show the performance of the WiFi-based sensing system when evaluating subjects with other conditions. Toward this goal, we additionally collect WiFi test data for patients with ALS (1 subject), MS (3 subjects), spinal cord injury (2 subjects), neurofibromatosis (1 subject), and dysferlinopathy (1 subject) amounting to a total of 8 subjects and 20 gait samples. It is worth emphasizing that these conditions are not seen during training or domain adaptation. We then feed these to our aforementioned classifier. The system was able to classify all these cases as unhealthy except for two

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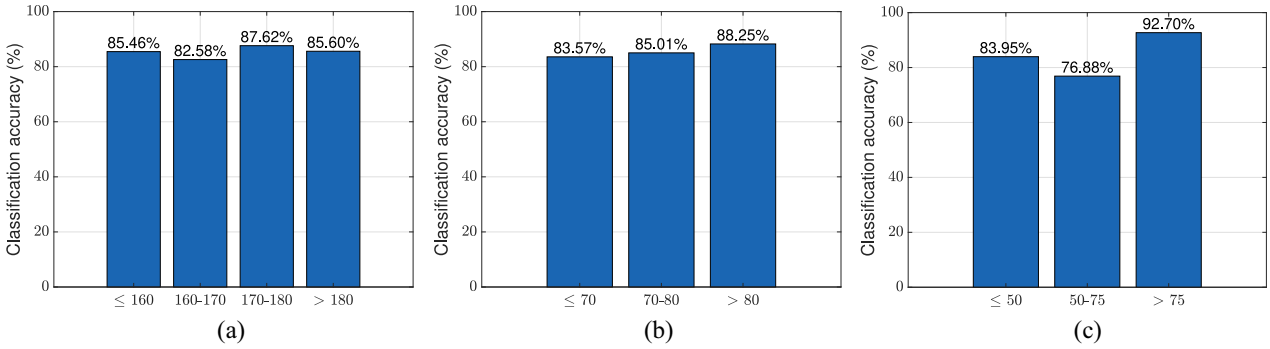


Fig. 6. Impact of (a) height, (b) weight, and (c) age on WiFi-based gait disorder assessment.

patients, one with ALS (always misclassified) and the other with neurofibromatosis (one of his/her sample walks was accurately classified in 2 out of the eight random samplings of domain adaptation-test selections while the other was always misclassified).

Overall, this shows the potential of the system to extend to unseen conditions. To improve the accuracy, sample data of these cases can also be included during training and/or domain adaptation.

E. Cases With Multiple Conditions

Some patients may suffer from multiple gait-related conditions. It is thus important to evaluate the performance of our classifier for such cases. Toward this goal, we have further recruited five patients who suffer from multiple conditions: arthritis and neuropathy (1 patient), dementia and post-stroke (1 patient), Parkinson's and neuropathy (1 patient), Parkinson's and Dementia (1 patient), arthritis, neuropathy and spinal cord injury (1 patient) amounting to a total of 10 gait samples.

All these patients were classified as unhealthy by our classifier except for the person with Parkinson's and neuropathy (one sample walk is accurately classified in half of the domain adaption-test selections while the other is accurately classified only in 1 out of 8 selections). Overall, this shows the robust performance of the system even in classifying cases with multiple conditions.

VII. Gait Disorder Classification Based on Videos

In this article, our goal is to provide a comprehensive investigation of gait disorder sensing with different modalities. Toward this goal, we next delve into developing a video-based gait disorder classifier. While there are a number of video-based gait disorder classifiers, as surveyed in Section II, the results of this section are important for the following reason. In order to effectively compare the performance of WiFi and video-based methods, it is crucial to test them under the same exact conditions (same subjects, environment, etc). In other words, they should be compared based on concurrent video and WiFi recordings of the same subjects. This is important since variability of the subjects can impact the severity of the underlying condition, and variability of the location/time can also result in different sensing qualities, preventing derivations of meaningful conclusions. We next lay out the details of our vision-based system and compare it with the previous WiFi-based pipeline.

Training Dataset: We use the same exact publicly available online video dataset, used for generating synthetic RF data in Section V-G, for training the vision-based system.

Test Subjects: For the purpose of testing, we need to concurrently collect videos of the same subjects that are used for testing the WiFi pipeline. The patient recruitment process is as described in Section V-D, where the patients have the option of participating in only WiFi data collection, or both video and WiFi. For those who agree to both, we then collect their video and WiFi simultaneously, with the video being recorded from the side since it is the most informative for mesh recovery (all online videos are side-views as well). The videos are then recorded using an ultrawide lens setting, with an aperture of f/2.4 and a 120-degree field of view, at a frame rate of 30 frames/s. We note that the scattered WiFi signals from the tripod and camera are filtered during the DC removal phase, thus not affecting the WiFi pipeline. *Finally, we note that our IRB committee has also reviewed and approved the video collection process.*

Feature Extraction and Training: The parameters we have used as features in the vision-based system are the same as those utilized in our WiFi pipeline. More specifically, we use the middle 50% of each walk to exclude the acceleration and deceleration parts. We then generate the 3-D point cloud of the body for each frame, which we then partition into 14 parts (hands, legs, torso, etc.). The average position of the points associated with each part results in 14 key points. We obtain the average, minimum (10% percentile), and maximum (90% percentile) of the torso speed as our first three features. To extract the gait cycle, we analyze the vertical position curves of each foot separately and over time. We then filter out the frequencies higher than 4 Hz to remove noise (the cadence for a fast run is around 180 strides/minute, which gives a gait cycle frequency of 1.5 Hz). Subsequently, we determine the gait cycle by computing the time difference between the extrema (at least 0.25 s apart) on each curve. Finally, we use the average of two gait cycles as the overall gait cycle, which serves as our fourth feature. It is worth mentioning that if, due to the noise of HMR and lack of visibility of one foot, the algorithm cannot extract the gait cycle from one foot, we only consider the gait cycle of the other foot. Finally, the step

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length and torso speed variations are obtained similar to our WiFi pipeline.

Using these six features, we then train a fully connected neural network with two hidden layers, while keeping all the specifications identical to the WiFi system (see Section V-I). For training, we have the same exact subjects and associated walk samples as in the video-to-RF training dataset of Section V-G. We note that while we had 101 subjects for WiFi testing in Section VI-A, we only have consents from 72 of those subjects for video recording. As such, later in this section, we will also further report the performance of the WiFi system with only those subjects as well, for a fair comparison.

More specifically, we have the following breakdown over the 72 subjects: Parkinson's disease (12), dementia (8), post-stroke (11), arthritis (hip or knee, 4), neuropathy (6), and healthy (31), totaling 293 test gait samples (average of 49 gait samples per condition). Among the patients (10 with mild, 14 with moderate, and 17 with severe gait disorders), 11 subjects (1 with a moderate disorder and 10 with severe disorders) used a walking aid, such as a cane. We note that we use both back and forth of the walks here as they both present informative side-views for a vision-based system. We then use the same number of patients/conditions for domain adaptation (12 total), as described for the WiFi pipeline. The overall performance is then averaged over 8 random sampling of the 60 subjects out of the overall pool of 72 subjects.

Performance Analysis: Fig. 7 summarizes the breakdown of the performance over different conditions, for our video-based classifier. Moreover, the overall average per subject accuracy of this classifier is 83.80%, with an average per class accuracy of 84.37%, with the following breakdown over the two classes: unhealthy (81.09%) and healthy (87.64%). As explained for the WiFi case, the variability of the performance among the conditions can be mainly attributed to the disparate distribution of disease severity across different conditions, or to different gait manifestation of the conditions. Overall, a video-based system can also be a key player in an automated gait disorder assessment system. However, it can raise privacy concerns or may not be favorable for some patients, motivating the use of RF signals.

WiFi-Based Versus Video-Based: Fig. 5 and 7 show the results of WiFi and video-based, respectively. However, as mentioned earlier, we did not have consents of all the 101 subjects for video recording. As such, while the two systems are trained on the same exact online video subjects, our video pipeline thus far is tested on 72 subjects while the WiFi is tested on a bigger set of 101 subjects. For a fair comparison, here we report the performance of the WiFi system when only tested on the same exact 72 subjects, while ensuring the same random sampling and domain-adaptation process. This results in an average per class accuracy of 84.41%, unhealthy (83.03%) and healthy (85.78%), with an overall average per subject accuracy of 84.19%. We can see that this does not impact the performance of the WiFi system, as compared to the results of Section VI-A, since the number of subjects is high. When comparing this WiFi's performance with that of the video, we can see that they are comparable. This is promising as it indicates the potential of WiFi for capturing gait disorders.

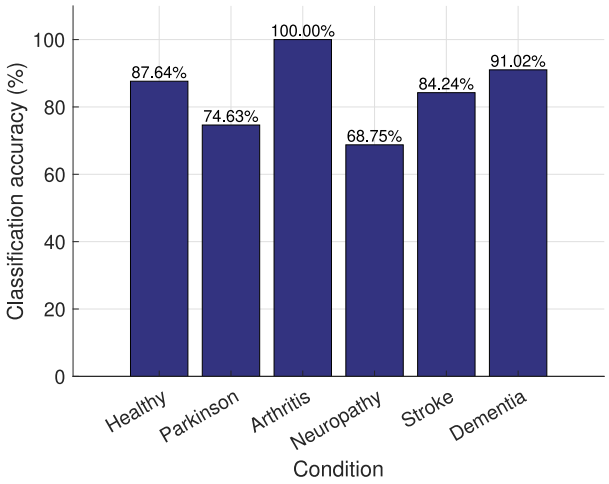


Fig. 7. Performance of our video-based system, when tested extensively with real patients exhibiting a spectrum of gait disorders, with an overall average per subject accuracy of 83.80%.

TABLE III
NEUROLOGISTS' ACCURACY WHEN DIAGNOSING GAIT DISORDER BASED ONLY ON VISUAL INSPECTION

Class	Classification Accuracy
Unhealthy	81.37%
Healthy	65.95%
Average per class accuracy	73.66%

It also creates the potential for multimodality sensing, as we discuss in Section IX-C. *To the best of our knowledge this is the first result that compares the performance of WiFi and video when capturing gait disorders, using the same exact subjects/conditions*. Such an apples-to-apples comparison is important for establishing meaningful conclusions.

VIII. MEDICAL EXPERTS' ACCURACY BASED ON VISUAL INSPECTION

As mentioned in Section II, neurologists rely on many inputs, in addition to a visual inspection of the gait, to establish a diagnosis. In this article, we have thus far shown the performance of gait disorder classification based on the information gait induces in the WiFi signal (or in a video). It is thus fitting to find the performance of neurologists when diagnosing a condition based only on the visual inspection of the walk, for proper comparison. *It is worth noting that this is the first of such a study, to the best of our knowledge*.

Toward this goal, we have designed a survey in Qualtrics [56]. Our survey is comprehensive, going beyond binary classification of the gait, which we shall discuss extensively in the next section. In this section, we discuss a part of the survey that facilitates comparison with our developed WiFi and video-based systems. Consider 72 questions (12 per each condition on average). Each question shows a video of a patient's gait and requests a diagnosis by selecting their top three preferred options from a list of six conditions of interest to this article (the same choice can be made for the top three as well). The videos are from the same subjects used for testing our video-based system for an apples-to-apples comparison. We have further partnered with SHG, an entity that provides

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TABLE IV
NEUROLOGISTS' ACCURACY WHEN DIAGNOSING SOLELY BASED ON THE VISUAL INSPECTION OF THE GAIT, WITH AN AVERAGE PER CONDITION ACCURACY OF 30.89%

Condition	Top-1 Acc.	Top-2 Acc.	Top-3 Acc.
Arthritis	18.63%	39.83%	51.04%
Healthy	65.95%	73.12%	80.39%
Dementia	14.70%	30.58%	44.29%
Neuropathy	9.35%	23.17%	40.78%
Parkinson's	35.15%	43.00%	48.11%
Post-stroke	41.55%	51.2%	57.19%

"measurable healthcare expert opinions," via recruitment and other tools [23], who then disseminated our survey to their pool of certified neurologists. In total, we have received an average of 197 responses per condition from 70 board-certified practicing neurologists, for each of the six conditions of interest to this article (five gait disorders + healthy). To evaluate the performance of neurologists in diagnosing gait-related conditions, we calculate a binary classification accuracy (healthy versus unhealthy) similar to the proposed WiFi or video systems. For this purpose, an answer is accurate if a neurologist identifies a healthy (unhealthy) subject as healthy (unhealthy) in their first choice.

Table III shows the results for each class (healthy and unhealthy) and the average per-class accuracy. *It is interesting to observe that both WiFi and video-based systems exhibit slightly superior performance compared to that of neurologists, when they only use visual inspection*. This is important as it has implications for how WiFi or video-based systems can be incorporated into a smart health system. As discussed in Section II, neurologists utilize many inputs when establishing a medical diagnosis. Nevertheless, some of these inputs can be easily collected and integrated with a WiFi or video-based sensing system. Such a system can then provide healthcare for a larger population, and can collaborate with neurologists as needed.

IX. DISCUSSION AND FUTURE WORK

We next discuss potential future directions.

A. Can RF Signals Differentiate Neurological Disorders?

Thus far, we have developed a WiFi-based gait disorder assessment system that can classify the gait into healthy and unhealthy. The natural next step would be to see if such signals are capable of not only identifying an unhealthy gait but also categorizing it to the specific underlying disorder. While there are several work on WiFi signals classifying different activities, such activities typical have a sizable footprint on the received signal. Exploring the efficacy of these signals in classifying gait disorders thus remains a subject of future exploration. It is worth noting that there are work in the area of vision on classifying the underlying disorder. It would be interesting to see how an RF-based system compares, and further expand the scope of the vision-based systems.

B. Experts' Accuracy Based on Visual Inspection–Classification

As outlined earlier, we conducted an extensive survey to evaluate the diagnostic accuracy of neurologists when relying solely on visual gait inspection. In Section VIII, we then detailed its findings regarding the classification of gait as either unhealthy or healthy. Here, we discuss the results when neurologists classify gait into one of the six conditions of interest to this article. This is important as it provides a benchmark for future RF-based work that aims to further classify the gait disorder to its underlying condition. Table IV summarizes the results, with an overall average accuracy of 30.89% per condition. These initial results indicate the importance of the other inputs (discussed in Section V-E) that are part of the diagnostic process, which have the potential to be more easily collected, and integrated with an RF-based or a video-based sensing system. Finally, it is noteworthy that despite using the identical dataset as in Fig. 7, the conditions challenging for assessment differ when assessed by humans.

C. Other Improvements

In this article, we only used one sample walk instance of the subject. Using more walk samples or a longer walk duration can further improve the performance. Moreover, we have used only 6 features for training the network, for both our WiFi and video-based systems. Additionally, the same 6 features are used for both systems for consistency. As part of future work, more features can be utilized and further tailored to each sensing modality. In addition, we have used Born approximation to simulate wave interaction with the body and generate synthetic data from videos. As part of future work, more advanced wave simulators can be used [44], at the cost of an increase in computational complexity. Also, a multimodal WiFi and video-based system can be developed to increase the robustness. In this article, we utilized the Linux 802.11n CSI Tool [53] for estimating the CSI of different subcarriers, as discussed in Section V-F. Other tools, such as the Atheros CSI Tool [57] or Nexmon [58], or newer WiFi standards can also be employed as part of future work. In addition, using more samples of different gait disorders for training can also improve the quality. For instance, we have only used samples of two gait disorders (plus healthy) during training/domain adaptation. Finally, in terms of disease severity, we only considered *mild, moderate* and *severe* cases. While considerably challenging, considering *very mild* cases can also be another interesting research avenue.

D. Toward Equitable Healthcare

An important future direction would be to integrate additional medical inputs (e.g., patient intake information, blood tests, brain scans) with the collected RF (or video) sensing data. Given that some of these inputs can be collected more easily (do not require expertise of neurologists), a successful integration can hold promise for an automated smart health system, paving the way toward equitable healthcare.

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X. Conclusion

In this article, we reported on the findings of a one-year-long clinical trial for gait disorder assessment. More specifically, we developed the first WiFi-based gait disorder sensing system of its kind, distinguished by its scope of validation with a large and diverse patient cohort. To ensure generalizability, our system was mainly trained on synthetic RF data, converted from publicly available videos of gait disorders, via developing a video-to-RF pipeline. We further developed a vision-based gait disorder assessment system under the same exact conditions, and provided the first apples-to-apples comparison between the WiFi-based and video-based ones. We finally contrasted both systems with the accuracy of neurologists when basing evaluation solely on the visual inspection of the gait. Overall, the results can play a key role for the proper integration of these sensing modalities into medical practice.

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Alireza Parsay (Member, IEEE) received the B.S. degree in electrical engineering from Shahed University, Tehran, Iran, in 2015, and the M.S. degree in electrical engineering, with a focus on communication systems, from Sharif University of Technology, Tehran, in 2017. He is currently pursuing the Ph.D. degree with the Department of Electrical and Computer Engineering, University of California at Santa Barbara, Santa Barbara, CA, USA.

His research interests include wireless sensing, imaging, and communication systems.



Mert Torun (Member, IEEE) received the B.S. degree in electrical and electronics engineering from Boğaziçi University, Istanbul, Türkiye, in 2018, and the M.S. degree in electrical and computer engineering from the University of California at Santa Barbara, Santa Barbara, CA, USA, in 2021, where he is currently pursuing the Ph.D. degree in electrical and computer engineering, in the area of communications, control, and signal processing.

His research interests include RF sensing, signal processing, computer vision, and machine learning.



Philip R. Delio received the B.S. degree in biology from Stanford University, Stanford, CA, USA, in 1987, and the M.D. degree from Georgetown Medical School, Washington, DC, USA, in 1995.

He completed a neurology residency and fellowship in cerebrovascular medicine in 2001. He currently serves as the Medical Director of the Comprehensive Stroke Center with Santa Barbara Cottage Hospital, Santa Barbara, CA, USA, where he is the past Chief-of-Staff. He also holds an adjunct professorship title with the University of California at Santa Barbara, Santa Barbara, for research in neurology and psychology.



Yasamin Mostofi (Fellow, IEEE) received the B.S. degree in electrical engineering from Sharif University of Technology, Tehran, Iran, in 1997, and the M.S. and Ph.D. degrees from Stanford University, Stanford, CA, USA, in 1999 and 2004, respectively.

She is currently a Professor with the Department of Electrical and Computer Engineering, University of California at Santa Barbara, Santa Barbara, CA, USA. Her research is multidisciplinary, in the areas of wireless systems and robotics/control. Her research has appeared in several reputable news venues, such as BBC and New Scientist. Her current high-level research thrusts include RF sensing for several different applications, such as through-wall imaging, occupancy analytics, smart health, and smart spaces; and communication-aware robotics, UAV-assisted connectivity, and joint robotic path planning and communication.

Prof. Mostofi is the recipient of the Presidential Early Career Award for Scientists and Engineers, the 2016 Antonio Ruberti Prize from the IEEE Control Systems Society, the National Science Foundation CAREER Award, and the IEEE 2012 Outstanding Engineer Award of Region 6 (more than ten Western U.S. states), among other awards. She was a semi-plenary speaker at the 2018 IEEE Conference on Decision and Control, a keynote speaker at the 2018 Mediterranean Conference on Control and Automation, and a keynote speaker at the 2023 IEEE International Symposium on a World of Wireless, Mobile & Multimedia Networks. She has served in many different professional capacities over the years. Her recent samples include serving on the inaugural editorial team for Nature NPJ Wireless Technology, serving as a Program Co-Chair for ACM MobiCom 2022, serving on the Board of Governors of IEEE CSS, and serving as a Senior Editor for IEEE TCNS.

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